A later cyclicity theorem for the open direction in arXiv:2011.04296

Literature-status note for `fa_banach_001`

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Status. `literature_already_answered (bounded peripheral-cyclicity regime)`.

Original open direction

In its contributions discussion, arXiv:2011.04296 says that it is unknown whether the generator of a uniformly eventually positive semigroup has cyclic peripheral spectrum under the same general conditions as in the positive case [1, p. 2]. This missing cyclicity theorem forces the paper to use a single-operator spectral-mapping workaround in its convergence argument.

For a generator A with spectral bound $\mathrm{s}(A)$, cyclicity means

$$\mathrm{s}(A) + i\beta \in \mathrm{spec}(A) \quad \Longrightarrow \quad \mathrm{s}(A) + in\beta \in \mathrm{spec}(A) \quad \text{for every } n \in \mathbb{Z}.$$

Later answer in the bounded regime

Theorem 4.7 of arXiv:2405.16371v2, on PDF page 12, proves exactly this cyclicity conclusion for every uniformly asymptotically positive C_0-semigroup on a complex Banach lattice [2]. The definition on PDF page 8 requires the rescaled semigroup to be bounded and requires that, uniformly for positive unit inputs, its distance from the positive cone tends to zero.

Consequently, the theorem applies in particular to every bounded uniformly eventually positive semigroup: after the common eventual-positivity time, the distance from the positive cone is identically zero. This gives the missing peripheral cyclicity theorem in the bounded regime that underlies the source's convergence discussion.

The later paper also closes the loop to convergence. Corollary 4.8 on PDF page 13 combines cyclicity with norm continuity at infinity to conclude that $\mathrm{s}(A)$ is dominant. The immediately following remark explicitly notes that Theorems 3.1 and 5.1 and Corollary 3.2 of the original paper remain valid when uniform eventual positivity is weakened to uniform asymptotic positivity. Thus the answering author knowingly applies the new theorem to the original convergence results, rather than the relation being a title-only match.

Scope limitations

This answer is intentionally scoped to bounded (after spectral rescaling) uniform eventual positivity, or more generally uniform asymptotic positivity. It does not claim to resolve the separate question in Remark 3.4 of the source about merely individually eventually positive semigroups. Nor does it address Remark 4.2(b), which asks whether the first-order-pole assumption can be removed from the source's Niiro–Sawashima theorem. This packet makes no new mathematical claim.

References

[1] S. Arora and J. Glück, *Spectrum and convergence of eventually positive operator semigroups*, Semigroup Forum **103** (2021), 791–811; arXiv:2011.04296, https://arxiv.org/abs/2011.04296.

[2] S. Arora, *Eventually positive semigroups: spectral and asymptotic analysis*, arXiv:2405.16371v2, https://arxiv.org/abs/2405.16371.